

Multiplicity and Basic Eigenvectors

Definition

A **basic eigenvector** of an $n \times n$ matrix A is any nonzero multiple of a basic solution to $(\lambda I - A)\vec{x} = 0$, where λ is an eigenvalue of A .

Problem

Find the eigenvalues and eigenvectors of $A = \begin{bmatrix} 4 & 1 & 2 \\ 0 & 3 & -2 \\ 0 & -1 & 2 \end{bmatrix}$, and $B = \begin{bmatrix} 3 & -4 & 2 \\ 1 & -2 & 2 \\ 1 & -5 & 5 \end{bmatrix}$.

$$\begin{aligned} \det(\lambda I - A) &= \det \begin{bmatrix} \lambda-4 & -1 & -2 \\ 0 & \lambda-3 & 2 \\ 0 & 1 & \lambda-2 \end{bmatrix} = (\lambda-4) \begin{vmatrix} \lambda-3 & 2 \\ 1 & \lambda-2 \end{vmatrix} = (\lambda-4) [(\lambda-3)(\lambda-2) - 2] \\ &= (\lambda-4) (\lambda^2 - 5\lambda + 4) \\ &= (\lambda-4)^2 (\lambda-1) \end{aligned}$$

we say that $\lambda=4$ has multiplicity 2.

For $\lambda_1 = 4$

$$(4I - A)\vec{x} = \vec{0}$$

$$\left[\begin{array}{ccc|c} 0 & -1 & -2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \vec{x} = \begin{bmatrix} s \\ -2t \\ t \end{bmatrix} = s \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

Hence, the eigenvalue $\lambda_1=4$ has two corresponding basic eigenvectors $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$.

Note that the basic eigenvectors are NOT scalar multiples of each other.

For $\lambda_2=1$

$$\left[\begin{array}{ccc|c} -3 & -1 & -2 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} -3 & -1 & -2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \vec{x} = \begin{bmatrix} -t \\ t \\ t \end{bmatrix} = t \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

In summary, A has two eigenvalues $\lambda_1=4$ and $\lambda_2=1$ with multiplicities 1 and 2 respectively. There are two basic eigenvectors corresponding to $\lambda_1=4$ and one basic eigenvector corresponding to $\lambda_2=1$.

Note that any linear combination of basic eigenvectors corresponding to the same eigenvalue is also an eigenvector of A . (See below)

$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$ are basic eigenvectors of A and so is $2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$.

$$\begin{aligned} A \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} &= 4 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \\ A \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} &= 4 \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} \end{aligned} \rightarrow A \left(2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} \right) = 4 \left(2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} \right)$$

$$B = \begin{bmatrix} 3 & -4 & 2 \\ 1 & -2 & 2 \\ 1 & -5 & 5 \end{bmatrix}$$

$$\lambda I - B = \begin{bmatrix} \lambda-3 & 4 & -2 \\ -1 & \lambda+2 & -2 \\ -1 & 5 & \lambda-5 \end{bmatrix}$$

$$\det(\lambda I - B) = \det \begin{bmatrix} \lambda-3 & 4 & -2 \\ -1 & \lambda+2 & -2 \\ -1 & 5 & \lambda-5 \end{bmatrix} = \det \begin{bmatrix} \lambda-3 & 4 & 2 \\ -1 & \lambda+2 & \lambda \\ -1 & 5 & \lambda \end{bmatrix} = \det \begin{bmatrix} \lambda-3 & 4 & 2 \\ 0 & \lambda-3 & 0 \\ -1 & 5 & \lambda \end{bmatrix}$$

$$= (\lambda-3)[(\lambda-3)\lambda+2] = (\lambda-3)(\lambda^2-3\lambda+2) = (\lambda-3)(\lambda-1)(\lambda-2)$$

$\lambda_1=1$, $\lambda_2=2$, $\lambda_3=3$ (They all have multiplicity one.)

Finding the eigenvectors:

$$\lambda_1=1 \rightarrow (I-B)\vec{x} = \vec{0} \rightarrow \left[\begin{array}{ccc|c} -2 & 4 & -2 & 0 \\ -1 & 3 & -2 & 0 \\ -1 & 5 & -4 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \vec{x} = \begin{bmatrix} t \\ t \\ t \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\lambda_2=2 \rightarrow (2I-B)\vec{x} = \vec{0} \rightarrow \left[\begin{array}{ccc|c} -1 & 4 & -2 & 0 \\ -1 & 4 & -2 & 0 \\ -1 & 5 & -3 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & -5 & 3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \vec{x} = \begin{bmatrix} 2t \\ t \\ t \end{bmatrix} = t \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

$$\lambda_3=3 \rightarrow (3I-B)\vec{x} = \vec{0} \rightarrow \left[\begin{array}{ccc|c} 0 & 4 & -2 & 0 \\ -1 & 5 & -2 & 0 \\ -1 & 5 & -2 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & -5 & 2 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \vec{x} = \begin{bmatrix} \frac{t}{2} \\ \frac{t}{2} \\ t \end{bmatrix} = \frac{t}{2} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$, and $\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$ are basic eigenvectors corresponding to $\lambda_1=1$, $\lambda_2=2$, and $\lambda_3=3$.

Note that if an $n \times n$ matrix A has m distinct basic eigenvectors $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_m$. As we have seen in the previous examples, the set $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_m\}$ is a linearly independent set (this follows from the definition of a basic eigenvector.) Hence, $\text{Span}\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_m\}$ is a subspace. In fact, since the vectors $\vec{x}_i$ ($1 \leq i \leq m$) are solutions of $(\lambda I - A)\vec{x} = \vec{0}$, we know that

$$\text{Span}\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_m\} = \text{Nul}(\lambda I - A),$$

Since this set is a subspace of $\mathbb{R}^n$, it is known as the **eigenspace of A** .

Theorem: Let A be an $n \times n$ matrix. Then, A is diagonalizable iff the matrix A has n linearly independent eigenvectors.

Proof: Assume that A has n linearly indep. eigenvectors $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n$. Then the matrix

$P = [\vec{x}_1 \ \vec{x}_2 \ \dots \ \vec{x}_n]$ is invertible and we have:

$$\begin{aligned} P^{-1}AP &= P^{-1}[A\vec{x}_1 \ A\vec{x}_2 \ \dots \ A\vec{x}_n] = P^{-1}[\lambda_1\vec{x}_1 \ \lambda_2\vec{x}_2 \ \dots \ \lambda_n\vec{x}_n] = P^{-1}[\vec{x}_1 \ \vec{x}_2 \ \dots \ \vec{x}_n] \underbrace{\begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_n \end{bmatrix}}_D \\ &= P^{-1}PD = D \end{aligned}$$

and hence A is diagonalizable.

Conversely, if A is diagonalizable, then there exists an invertible matrix P such that $D = P^{-1}AP$ is diagonal, hence $AP = PD$. Now assume $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n$ are columns of P , then

$$P = \begin{bmatrix} \vec{x}_1 & \vec{x}_2 & \dots & \vec{x}_n \end{bmatrix} \longrightarrow \text{For } i=1,2,\dots,n: A\vec{x}_i = \lambda_i \vec{x}_i$$

which means the diagonal entries of D are the eigenvalues of A and $\vec{x}_i$'s are their corresponding basic eigenvectors which are linearly independent (since we know P is invertible).

Revise the previous example: Note that A is 3×3 and has 3 distinct basic eigenvectors

As a result, we saw that A is diagonalizable.

Also, notice that B has 3 distinct eigenvalues and each eigenvalue has a basic eigenvector corresponding to it, hence B is diagonalizable as well.

Characterizing diagonalizable matrices

Theorem

A square matrix A is diagonalizable if and only if every eigenvalue λ of multiplicity m yields exactly m basic eigenvectors, i.e., the solution to $(\lambda I - A)X = 0$ has m parameters.

Remark

A special case of this result is when every eigenvalue of a matrix has multiplicity one. An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

Similar Matrices

Definition

Let A , and B be $n \times n$ matrices and suppose there exists an invertible matrix P such that $A = P^{-1}BP$. Then A and B are called **similar matrices**.

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix} \quad P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad P^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

A and B are similar matrices.

$$P^{-1}BP = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix} = A$$

observation: P^{-1} is obtained from I by switching the columns: $\begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$ Columns are Switched

Theorem

Let A and B be similar matrices such that $A = P^{-1}BP$, where A, B are $n \times n$ matrices and P is invertible. Then A and B have the same eigenvalues.

$$P^{-1}P = I$$

$$\begin{aligned} \det(\lambda I - A) &= \det(\lambda I - P^{-1}BP) = \det(P^{-1}\lambda I P - P^{-1}BP) \\ &= \det(P^{-1}(\lambda I - B)P) = \det(P^{-1}) \det(\lambda I - B) \det(P) = \frac{1}{\det(P)} \det(\lambda I - B) \det(P) = \det(\lambda I - B) \end{aligned}$$

Hence A and B have the same characteristic polynomial and therefore the same eigenvalues.

In particular if A is diagonalizable, then A is similar to a diagonal matrix D and hence the eigenvalues of A and D are the same, which means the diagonal entries of D are the eigenvalues of A .

Problem

Use the previous theorem to find the eigenvalues of the following matrix.

$$A = \begin{bmatrix} 33 & 105 & 105 \\ 10 & 28 & 30 \\ -20 & -60 & -62 \end{bmatrix}$$

Note that

$$\begin{bmatrix} 33 & 105 & 105 \\ 10 & 28 & 30 \\ -20 & -60 & -62 \end{bmatrix} \xrightarrow{2R_2 + R_3} \begin{bmatrix} 33 & 105 & 105 \\ 10 & 28 & 30 \\ 0 & -4 & -2 \end{bmatrix} \xrightarrow{-2C_3 + C_2} B = \begin{bmatrix} 33 & -105 & 105 \\ 10 & -32 & 30 \\ 0 & 0 & -2 \end{bmatrix}$$

$\downarrow$ left multiply by E $\downarrow$ right multiply by E^{-1}

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix} \qquad E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

$$B = EAE^{-1} \quad \text{or} \quad A = E^{-1}BE \quad \longrightarrow \quad A \text{ and } B \text{ are similar matrices and have the same eigenvalues.}$$

$$\begin{aligned} \det(\lambda I - A) &= \det(\lambda I - B) = (-1)^3 \det(B - \lambda I) = -\det \begin{bmatrix} 33-\lambda & -105 & 105 \\ 10 & -32-\lambda & 30 \\ 0 & 0 & -2-\lambda \end{bmatrix} \\ &= -(-2-\lambda) \begin{vmatrix} 33-\lambda & -105 \\ 10 & -32-\lambda \end{vmatrix} \\ &= \dots \end{aligned}$$

Theorem (another addition to invertible matrix theorem)

Let A be an $n \times n$ matrix. Then A is invertible iff 0 is not one of its eigenvalues.

We know that A is invertible iff $\det A \neq 0$. To prove the above theorem, we need to show that $\det A \neq 0$ iff 0 is not one of its eigenvalues.

We first show that $\det A = \lambda_1 \lambda_2 \dots \lambda_n$, where λ_i 's are the eigenvalues of A .

Remember that λ_i 's are roots of the characteristic polynomial, hence:

$$\det(\lambda I - A) = (\lambda - \lambda_1)(\lambda - \lambda_2) \dots (\lambda - \lambda_n)$$

Now, choose $\lambda = 0$, then:

$$\det(-A) = (-1)^n \lambda_1 \lambda_2 \dots \lambda_n$$

Also, we know $\det(-A) = (-1)^n \det A$. Therefore:

$$\det A = \lambda_1 \lambda_2 \dots \lambda_n.$$

Finally, it is clear if one of the eigenvalues is zero, then $\det A = 0$ and if $\det A = 0$ then at least one of the eigenvalues must be zero.